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B.Tech. Degree VII Semester Special Supplementary Examination April 2025

ME 19-205-0705 SUPPLY CHAIN MANAGEMENT (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Define a supply chain and understand its different structures and its importance to the success of a firm.
 CO2: Understand the concept of strategic fit between the SC strategy and the competitive strategy of the firm and how to achieve it.
 CO3: Identify the main drivers of supply chain (SC) performance and measure them using precise metrics.
 CO4: Assess the importance of distribution networks and the different opinions available for their design.
 CO5: Develop a methodological framework for network design and facility location and use optimisation models and techniques for facility location and capacity allocation.
 CO6: Use safety inventory to deal with demand uncertainty in the supply chain.
 CO7: Understand the importance of transportation in the SC, identify the different modes of transportation and the role played by the infrastructure and policies.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome

PART A (Answer ALL questions)

2023

	(8 × 3 = 24)	Marks	BL	CO	PO
(a) What are the key components or stages of a typical supply chain?	3	L1	1	1	
(b) Explain the role of logistics in supply chain management.	3	L2	2	1	
(c) Explain the concept of ordering cost and carrying cost in inventory management in supply chain.	3	L2	3	1	
(d) How safety stock is calculated to prevent stock-outs in supply chain?	3	L3	6	1	
(e) What role does transportation play in the overall efficiency of the supply chain?	3	L3	4	2	
(f) Explain the concept of inter-modal transportation in modern supply chains.	3	L2	7	1	
(g) What factors influence the decision to select a facility location in supply chain network?	3	L3	5	1	
(h) How can companies optimize capacity allocation to minimize costs while meeting customer demand?	3	L2	5	1	

PART B

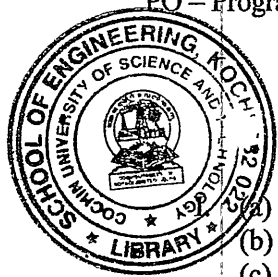
(4 × 12 = 48)

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|-----|-----|---|---|----|---|---|
| II. | (a) | Explain various supply chain decision phases. | 6 | L2 | 1 | 1 |
| | (b) | Explain why achieving strategic fit is critical to a company's overall success. | 6 | L3 | 2 | 2 |

OR

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|------|-----|---|---|----|---|---|
| III. | (a) | How has the implementation of GST (Goods and Services Tax) influenced supply chain operations and logistics performance in India? | 6 | L2 | 1 | 1 |
| | (b) | Explain how a supply chain function within a business. | 6 | L2 | 3 | 2 |

(P.T.O)



		Marks	BL	CO	PO
IV.	(a) Explain main types of inventory in a supply chain.	3	L2	5	1
	(b) How does the inventory turn over ratio help in assessing the efficiency of inventory management?	4	L2	5	2
	(c) A retail shop in Delhi sells 1500 units of a product weekly. The lead time to restock is 3 weeks and the standard deviation of weekly demand is 45 units. The desired service level is 95% and the Z-value for 95% is 1.65. Calculate the required safety stock.	5	L3	6	2
OR					
V.	(a) XYZ company has an average daily demand of 90 units. The lead time for orders varies with an average of 9 days and a standard deviation of 2 days. The company wants 98% service level with a Z-value of 2.33. What is the reorder point (ROP)?	7	L3	6	2
	(b) Explain the impact of inventory centralization on various supply chain performance measures.	5	L2	5	1
VI.	(a) What are the key challenges in managing transportation within global supply chains?	5	L2	4	1
	(b) What are the different modes of transportation used in supply chains and how do companies decide which to use?	7	L2	7	1
OR					
VII.	(a) What is cross-docking? What are the difficulties in implementing cross-docking?	6	L2	5	2
	(b) How does multi-modal transportation improve supply chain efficiency in India?	6	L2	5	2
VIII.	(a) What factors influence the decision to select a facility location in a supply chain network?	6	L2	5	4
	(b) What is capacity allocation and why is it important in supply chain management?	6	L2	5	4
OR					
IX.	(a) How does demand forecasting influence capacity allocation decisions in a supply chain?	6	L2	5	4
	(b) What are the challenges of balancing capacity allocation between different facilities or production lines in a supply chain?	6	L2	5	4

Bloom's Taxonomy Levels

L1 – 2.5%, L2 – 75%, L3 – 22.5%.
